

BBO capstone project presentation template

Develop a concise presentation that captures:

1. An overview of your BBO approach

- Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
- Briefly describe your overall strategy and how your process fits together.

Core Objective and Purpose

The overarching ambition of this capstone project is to discover the global maximum of eight unknown, computationally expensive mathematical benchmark functions defined across continuous unit hypercubes spanning two to eight dimensions. Because we are denied analytical equations, gradient vectors, and internal structural visibility, and are restricted to an exacting budget of strictly one evaluation per function per weekly round, our fundamental objective is to maximise sample efficiency by identifying optimal coordinate configurations with minimal query expenditure.

Overall Strategy and High-Level Narrative

Our framework operates as an automated, closed-loop Bayesian active learning pipeline that iteratively constructs probabilistic surrogate models of the hidden landscapes from empirical evaluation records. At each cycle, the pipeline executes rigorous five-fold cross-validation across diverse regressors (incorporating Automatic Relevance Determination Gaussian Processes, multi-layer perceptron neural networks, and decision tree ensembles) to select the most dependable surrogate. Query candidates are generated via a hybrid mechanism combining quasi-random Latin Hypercube exploration with localized perturbation envelopes around historical elite points, filtered through Euclidean distance de-duplication to prevent redundant queries, and chosen via Expected Improvement and Upper Confidence Bound acquisition criteria.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

Key Changes in Approach Since Early Rounds

Our methodology has evolved from tentative, uninformed quasi-random exploration into an anisotropic, variance-driven active learning system defined by mathematical discipline. In the initial rounds, when sparse data precluded reliable surrogate approximation, our workflow relied on broad Latin Hypercube sweeps governed by uniform spatial priors and substantial epistemic uncertainty across all dimensions. As our empirical dataset expanded past twenty points per function, we discarded isotropic assumptions in favour of coordinate-specific sensitivity modeling, replacing unguided perturbations with structured active subspace exploitation.

Influencing Drivers (Data Trends, Feedback, and Performance)

This progression was driven by empirical evaluation histories, cross-validation diagnostic scores, and the pronounced curse of dimensionality encountered in moderate-to-high dimensions. Treating all coordinate axes identically diluted our sparse query budget across uninformative directions and caused early models to stagnate against boundary walls. Rigorous cross-validation benchmarks demonstrated that well-calibrated Gaussian Processes with anisotropic covariance kernels consistently outperformed unregularised deep regressors in low-sample regimes, prompting us to anchor our decisions to probabilistic lengthscale telemetry.

Guiding Principles and Operational Heuristics

Our query decisions are now steered by four core heuristics:

Automatic Relevance Determination to isolate sensitive active subspaces from invariant coordinate dimensions.

Mandatory Euclidean distance de-duplication enforcing a minimum separation of 0.005 from all historical evaluations to preserve novel informational gain.
Adaptive acquisition fallbacks that transition smoothly from Expected Improvement to Upper Confidence Bound whenever improvement gradients diminish near saturated regions.

Multi-model predictive calibration that prioritises surrogates with superior cross-validation generalisation over raw training accuracy.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

Meaningful Trends Across Data Points and Evaluations

Analysing our cumulative dataset across 263 evaluations revealed that objective response landscapes are rarely uniform; rather, they partition into low-dimensional active manifolds, steep localized ridges, and narrow ascending basins nestled within broader penalty plateaus. This structural insight underpinned our breakthrough twelfth round, where focused exploitation unlocked three brand-new all-time record highs: Function 6 ascending to -0.202865, Function 7 surging to an unprecedented 2.661908, and Function 8 achieving an elite peak of 9.966558.

Variables and Behaviours Driving Results

Performance variation is overwhelmingly dictated by low-dimensional principal coordinate combinations rather than full nominal dimensional spaces. In Function 8, spanning eight dimensions, variation is dominated by an active subspace where coordinates one and two collapse towards zero while coordinate five nears unity, governing our path to the 9.966558 maximum. Similarly, Function 5 displays an extreme directional gradient that elevates objective values by orders of magnitude (from 171 to 8662) as coordinates simultaneously approach the upper vertex (1.0, 1.0, 1.0, 1.0), whereas Function 4 features a delicate positive centroid island (centred near 0.40, 0.42, 0.38, 0.34) surrounded by severe negative penalties.

Reshaping Our Understanding of the Search Process

These empirical observations have fundamentally reshaped our conceptual model of black-box optimisation, demonstrating that apparent high-dimensional complexity frequently masks much simpler intrinsic manifolds. By treating search spaces through the lens of coordinate relevance and identifying directional gradients, we transformed our search from blind spatial sampling into a principled ascent along dominant topological corridors.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Balancing Exploration and Exploitation

We govern the balance between exploration and exploitation through probabilistic acquisition functions that mathematically trade off predictive mean (exploitation) against epistemic posterior variance (exploration). In earlier rounds, broad uncertainty envelopes guided queries towards undersampled regions to uncover landscape contours; however, as iterations advanced and high-yield basins were confirmed, we systematically contracted candidate perturbation radii around elite clusters, utilising range-scaled Expected Improvement to concentrate search density where expected returns were highest.

Strategic Examples (What Worked vs What Did Not)

What Worked:

Isolating the active subspace in Function 8 and concentrating queries within the ascending basin of Function 7 produced consecutive record-breaking peaks (2.661908 and 9.966558), while targeting the extreme upper vertex in Function 5 unlocked our monumental 8662.482500 score.

What Did Not Work:

Early isotropic perturbations in high-dimensional spaces frequently dropped evaluations into catastrophic penalty valleys (such as Function 4 plunging below -36), and unscaled Expected Improvement occasionally caused candidate clustering along domain boundaries without probing interior maxima.

Handling Uncertainty and Adapting to Negative Surprises

When an evaluated query yields an unexpected drop or contradicts surrogate predictions, our framework treats the outcome as valuable epistemic evidence that refines our posterior belief. Rather than overreacting with chaotic parameter changes, the system incorporates the negative point to update covariance lengthscales, inflates predictive uncertainty across the surrounding neighbourhood to deter immediate re-sampling, and activates distance-constrained orthogonal candidates to explore alternate topographical hypotheses.

5. Next steps and reflection

- Outline your planned actions or experiments to improve the model's performance.
- Include two or three sentences connecting your work to the broader ML landscape.
- Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Planned Actions for the Final Round (Module 24)

With Module 24 representing the terminal query round of the entire capstone campaign, the future value of acquiring exploratory information drops to zero, rendering speculative variance-seeking irrational under finite-horizon decision theory. Consequently, our final submission strategy transitions decisively to pure exploitation: we will compress candidate perturbation envelopes tightly around our confirmed global peaks (specifically Functions 6, 7, and 8), reduce exploratory acquisition weights, and employ fine-grained numerical gradient climbing along dominant principal axes to extract the final increments of objective performance.

Broader Machine Learning Context

This sequential optimisation challenge directly mirrors fundamental real-world machine learning problems, including automated hyperparameter tuning for multi-billion-parameter foundation models, chemical molecular discovery, and physical robotic control systems where live trials are exceptionally hazardous or expensive. In all these critical applications, the core principles developed here (rigorous uncertainty quantification, sample-efficient active learning, and inductive bias selection) form the bedrock of modern artificial intelligence engineering.

Communicating Results to Non-Technical Stakeholders

"Imagine searching for the highest mountain peak in an unfamiliar landscape shrouded in dense fog, where every single step takes a full week and costs substantial resources; rather than wandering blindly through the mist, our intelligent framework uses a 'mathematical radar' to build an increasingly accurate virtual map from every footstep, systematically navigating through the terrain to guide us straight to the highest summit with maximum efficiency and minimum risk."